

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 412 — Past Questions & Materials
by Renance

Course: CHM 412
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200, 400

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Paper 1 of 2 — CHM 412 - 2024/2025 - First Semester - Coordination Chemistry

Level 200 • First Semester • PastQuestion

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Paper 2 of 2 — CHM 412 PQ 2

Level 400 • First Semester • PastQuestion

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Federal University of Petroleum Resources, Effurun
Department of Chemistry
B.Sc (Hons.) Degree Examination, First Semester 2016/2017 Session; March 2017
CHM 412- Coordination Chemistry (2 credits)
Answer any Three questions only. Time Allowed-2 hours

EF7 / CF+

a) Discuss the ligand field treatment of bonding in octahedral complexes of transition metals [10 marks].

b) Give any three industrial applications of chelates [3 marks].

c) Co^{3+} is stabilized in the presence of strong field ligands while Co^{2+} is stabilized in the presence of weak field ligands. Explain [5 marks].

d) Write short note on outer orbital complex [6 marks].

e) Predict the magnetic moment for $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ ions [6 marks].

f) Write short notes on the following:
(i) Jahn-Teller effect [7 marks].
(ii) The Irving-Williams series [5 marks].
(iii) High spin complex [6 marks].

g) Calculate the spectral term symbol for the following complexes:
(i) $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_3$ [2 marks]
(ii) $[\text{V}(\text{H}_2\text{O})_6]\text{Cl}_3$ [2 marks]
(iii) $[\text{Fe}(\text{NH}_3)_6]^{2+}$ [2 marks].

h) For each coordination number from two to seven, mention the principal geometrical arrangement (or arrangements) [6 marks].

i) What is the spectrochemical series [5 marks]?

j) Draw the d orbitals [3 marks].

k) State which d orbitals fall into the e_g set in an octahedral ligand field [2 marks].

l) State which d orbitals fall into the t_{2g} set in an octahedral ligand field [2 marks].

m) The compound of $\text{CoCl}_2 \cdot 4\text{NH}_3$ contains only one Cl^- ion that is precipitated immediately on the addition of Ag^+ ions. Draw the structure of the compound on the basis of Werner's coordination theory [5 marks].

n) Explain the concept of Werner's theory [8 marks].

o) SCN^- ion shows linkage isomerism in coordination compounds. Explain [3 marks].

p) Consider the reactions:
 $\text{Ni}^{2+}(\text{aq}) + 6\text{NH}_3 \rightleftharpoons [\text{Ni}(\text{NH}_3)_6]^{2+}(\text{aq}) \quad \log \beta = 8.60$
 $\text{Ni}^{2+}(\text{aq}) + 3\text{en} \rightleftharpoons [\text{Ni}(\text{en})_3]^{2+}(\text{aq}) \quad \log \beta = 18.20$
Account for the huge discrepancy in the values of β [6 marks].

Handwritten notes and calculations:

$16 + 8 = 24$

$12 + 1 + 3 = 16$

$10 + 6 + 18 = 34$

$4 + 1 = 5$

$2 + 1 + 0 = 3$

$3 - 3 = 0$

$k = 4$

$k = 5$

$k = 3$

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$k = 2$

$k = 3$

$k = 4$

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